

SUMMARY OF THE INVENTION

The present disclosure provides systems and methods for longitudinal monitoring of pelvic physiology using a body-positioned intraluminal physiologic monitoring device configured to acquire physiologic information across time while maintaining user comfort, privacy, and repeatable measurement conditions.

In various embodiments, the device comprises an insertable body configured for placement within a body lumen for temporary measurement sessions, periodic monitoring, retained placement, or repeated insertion events. The device may be configured for vaginal, anal, or other intraluminal environments without limitation to specific anatomical implementations.

The intraluminal monitoring device may include one or more sensing subsystems configured to detect physiologic conditions associated with tissue interaction, environmental context, and biologic response. Sensing modalities may include, without limitation:

- bio-impedance sensing including single-frequency or multi-frequency impedance spectroscopy;
- capacitive or dielectric sensing configured to infer hydration or moisture-related physiologic proxies;
- distributed pressure sensing, force sensing resistors, capacitive pressure elements, strain sensors, or compliance-detection structures;
- temperature sensing elements configured to capture thermal context and physiologic variability;
- chemical or electrochemical sensing including pH sensing or related proxy measurements;
- motion or orientation sensing using inertial measurement units for posture detection and motion artifact rejection;
- acoustic, vibrational, or mechanical response sensing;
- optical or electromagnetic sensing modalities; and
- combinations of the foregoing.

In certain embodiments, sensing elements may be distributed longitudinally along the insertable body to enable spatial characterization of tissue interaction, contact stability, compliance behavior, or physiologic gradients along the monitored region.

The device may further include internal electronics comprising one or more processors, signal conditioning circuitry, wireless communication interfaces, power storage components, and sealed protective housings configured for operation within biologic environments.

Physiologic signals collected by the device may be processed locally, transmitted to an external computing device, or analyzed within remote computational environments configured to derive longitudinal physiologic profiles. Derived outputs may include abstracted indices, trend curves, variability signatures, recovery patterns, contact stability metrics, hydration-related proxies, compliance indicators, or other non-explicit representations of physiologic behavior.

In various embodiments, the intraluminal monitoring device may operate independently or as a functional node within a distributed physiologic sensing ecosystem comprising multiple body-positioned sensing devices. Data fusion between nodes may enable contextual interpretation of physiologic states, cross-body correlation analysis, adaptive calibration, compatibility modeling, or personalized health intelligence generation.

Embodiments described herein are illustrative and non-limiting. The disclosure encompasses alternative form factors, sensor technologies, materials, electronic architectures, insertion geometries, and computational methods capable of achieving substantially similar functional outcomes.